

# Tunisia, audited — investor economics

The FX bridge by leg, per-pathway DSCR computed in code, the capital stack, and the exact signatures that move “commercially validated” from zero.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations\_audit.py / make\_nations\_pdfs.py — the same code that renders the page.

## 1. The FX bridge, tiered by confidence

| LEG                                           | \$M/YR            | GRADE                        |
|-----------------------------------------------|-------------------|------------------------------|
| Strategic metals (U + heavy REE, acid stream) | \$75-86M          | <b>STRONG</b>                |
| Solar fuel substitution (500 MW)              | \$95-116M         | <b>PLAUSIBLE</b>             |
| Hydrogen → ammonia (loop-internal offtake)    | \$7-25M           | <i>SPECULATIVE</i>           |
| Terfas (desert truffles)                      | \$14-29M          | <b>PLAUSIBLE</b>             |
| Food import substitution                      | \$30-30M          | <i>SPECULATIVE</i>           |
| Compute colocation services                   | \$15-15M          | <i>SPECULATIVE</i>           |
| ESG financing delta                           | \$8-12M           | <b>PLAUSIBLE</b>             |
| <b>Total (mid)</b>                            | <b>~\$279M/yr</b> | <b>tiered by grade below</b> |

\$80M carries a proven flowsheet or a cited anchor; \$137M is gated; \$61M rests on assumed volume or price. Zero dollars are commercially validated — the first buyer moves the ledger.

### 1.1 The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **\$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **\$9.34B** = 4.3 months of imports; food \$3.00B + fuel \$4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

**Channel 1 — the debt trap:** the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

**Channel 2 — imported inflation:** the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food (\$3.00B) and fuel (\$4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

**The loop’s answer:** owned export engines earn ~\$117M/yr of gross exports outside the Fed cycle; the substitution legs take \$152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 →

BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

**The macro context is the thesis:** the same Fed transmission that sets the emerging-market cycle prices this state's debt, and the loop's engines are the only line item in the country that earns hard currency *outside* that channel. That is why the FX bridge above is the investor's first table, not an appendix.

## 2. Pathway economics — computed in code (nations\_v5.py)

### P2 • Strategic metals (acid-stream U + heavy REE)

| LINE                    | VALUE                                                           |
|-------------------------|-----------------------------------------------------------------|
| Revenue                 | \$75-86M/yr (716.5k lbs U&sb3;O&sb8; @ \$80-95/lb + ~585 t REO) |
| OPEX                    | \$54-74M/yr (SX \$59-83/lb + shared circuit)                    |
| Gross margin            | 0-37%                                                           |
| CAPEX / payback         | ~\$100M / 6.3 yr (mid)                                          |
| DSCR (60% debt @7%/15y) | 2.40 — passes at contract prices                                |
| Gate                    | pilot U recovery ≥85% · all-in ≤\$75/lb · offtake signed        |

### P3 • Solar IPP (500 MW) — the honest finding

| LINE                         | VALUE                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------|
| Revenue / OPEX               | \$27.2M / \$8M (500 MW × 20% CF × \$31/MWh)                                                       |
| CAPEX                        | ~\$550M                                                                                           |
| DSCR @8%/20y                 | <b>0.46</b> — fails                                                                               |
| DSCR @4.5% DFI/25y           | <b>0.69</b> — fails                                                                               |
| Bankable region (DSCR ≥1.15) | \$0.70/W + 21% CF · \$0.70/W + 22% CF · \$0.70/W + 23% CF · \$0.70/W + 24% CF · \$0.80/W + 24% CF |
| Clearing PPA                 | ≥\$45652/MWh at mid-case capex/CF                                                                 |
| Verdict                      | bankable ONLY lean, inside the region, or with grant/sovereign support — published, not hidden    |

### P4 • Municipal desalination (150,000 m<sup>3</sup>/d)

| LINE         | VALUE                                       |
|--------------|---------------------------------------------|
| Revenue      | \$30.1M @\$0.55 → \$38.3M @\$0.70 blended   |
| OPEX / CAPEX | \$24.6M (@\$0.45/m <sup>3</sup> ) / ~\$165M |
| DSCR @\$0.55 | <b>0.54</b> — fails                         |
| DSCR @\$0.70 | <b>1.36</b> — clears                        |

| LINE | VALUE                                                                   |
|------|-------------------------------------------------------------------------|
| Gate | ≥60% contracted offtake · blended ≥\$0.68/m <sup>3</sup> · brine permit |

### 3. The capital stack and the ask

**Phase o:** \$150-250k, 90 days, no construction — the price of replacing every projection with a measurement (itemized in the complete audit). **Then, module by module, on public gates:** \$58k health pilot · \$100M metals · \$550M solar · \$165M desal · \$5-30M truffles · \$2-5M food. Security per module: offtake assignment, STEG PPA, municipal contracts, MIGA/DFI channel — modeled, not committed.

#### What moves “commercially validated” from zero

GCT MOU + first assay batch · STEG PPA award · desal LOI ≥60% at ≥\$0.68/m<sup>3</sup> · first truffle fruit + ≥€20/kg contract · 100-point water panel + 90-day health KPIs · 3,000-family cohort baseline. Each is a Phase-o/1 deliverable with a date.

**What this document is:** a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

**What it is not:** a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

#### Works cited

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- [Ru2] URANEXT/OCP (Morocco) building the first modern yellowcake-from-phosphoric-acid plant, ~\$100M CAPEX, El Jadida, 2025 — <https://northafricapost.com/87963-uranext-to-build-moroccos-first-yellowcake-uranium-plant-in-el-jadida.html...>
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